

Dynamic Forecasting of Stock Market Returns: Comparative Analysis of ARIMA-GARCH & LSTM Models

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Abstract

This investigation delves into the comparative dynamics of stock market forecasting, focusing on the efficacy of the ARIMA-GARCH and Long Short-Term Memory (LSTM) neural network models. Our analysis encompasses a decade-long dataset from January 2013 to December 2023, including 20 leading stocks like AAPL, MSFT, AMZN, and GOOGL. An initial examination of the data's stationarity confirmed its compatibility for sophisticated time-series modeling. The ARIMA-GARCH model, renowned for adeptly managing series with intrinsic volatility patterns, was meticulously adapted for each stock, with its predictive accuracy appraised via Mean Squared Error (MSE) and Mean Absolute Error (MAE). Concurrently, the LSTM model's potential to encapsulate prolonged data dependencies was rigorously tested. Although both models demonstrated formidable predictive power, our findings suggest a nuanced superiority of the ARIMA-GARCH model in minimizing prediction errors, as quantified by MAE. These insights underscore the enduring pertinence of established econometric techniques amidst evolving machine learning paradigms.

Keywords – ARIMA-GARCH, LSTM, Neural Network Models, MSE, MAE

I. Introduction

The financial market is a complex adaptive system characterized by high volatility and noise, posing significant challenges to accurate forecasting. Predictive modeling in finance has long been the crux of quantitative analysis, where the ability to anticipate market behavior can yield substantial rewards. This paper focuses on the intricate task of forecasting stock market returns and volatility, a pivotal concern for investors, analysts, and policymakers. Traditional econometric models, particularly the Autoregressive Integrated Moving Average (ARIMA) coupled with the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) approach, have stood the test of time, offering robust performance in volatility estimation and return prediction. Concurrently, the emergence of deep learning techniques has ushered in a new era of predictive analytics, with Long Short-Term Memory (LSTM) networks demonstrating promising results in modeling sequential and time-series data.

The decade of the 2020s has emerged as a crucible for financial models, where the predictive acumen of ARIMA-GARCH and LSTM is rigorously tested against a backdrop of market anomalies and extremes. The onset of the global COVID-19 pandemic served as a catalyst for profound volatility, revealing the susceptibility of global markets to systemic shocks. As the

pandemic unfolded, financial markets oscillated between precipitous declines and rapid ascents, a testament to the unprecedented levels of uncertainty that investors and models alike had to navigate.

This period of upheaval was characterized by frenetic trading activity, as algorithmic systems responded to real-time news with heightened sensitivity, amplifying market movements. The ARIMA-GARCH model, with its inherent design to account for volatility clustering and time-varying volatility, was poised to adapt to these conditions, offering a structured approach to quantify risk and return in turbulent times. On the other hand, the LSTM model, with its deep learning capabilities, sought to understand and leverage the complex patterns emerging from the market's reactions to a confluence of socio-economic disruptions.

Moreover, the pandemic's economic ramifications prompted unprecedented monetary interventions. Central banks across the globe embarked on expansive monetary easing, injecting liquidity to stabilize markets and fuel economic recovery. This era of low interest rates and expansive fiscal policy created a dichotomy in the markets, with asset price inflation contrasting economic reality, challenging models to differentiate between liquidity-driven growth and fundamental economic strength.

The retail trading boom, fueled by the democratization of trading through technology and spurred by increased savings and disposable income during lockdowns, further complicated market dynamics. The phenomenon of 'meme stocks', driven by social media and retail investor consortiums, defied traditional valuation metrics, creating pockets of the market that operated on sentiment rather than fundamentals, presenting a novel challenge for predictive models that traditionally relied on historical data and rational market behavior assumptions.

In this comparative study, we systematically analyze and benchmark the performance of the ARIMA-GARCH framework against the LSTM model across 20 diverse stocks over a decade-long period (January 2013 to December 2023). The investigation begins with a rigorous stationarity test to confirm the suitability of time-series data for the application of ARIMA-based models. Subsequently, we delve into the intricacies of model building, parameter tuning, and evaluation, employing Mean Squared Error (MSE) and Mean Absolute Error (MAE) as metrics to assess the accuracy of return forecasts. The findings of this study not only shed light on the efficacy of each modeling approach in capturing the dynamics of stock returns but also provide empirical evidence on their performance nuances, thereby guiding strategic investment decisions and reinforcing the foundation for future financial modeling innovations.

This work aligns with the scientific pursuit of enhancing forecasting techniques amidst the evolving landscape of machine learning and financial econometrics, reaffirming the necessity for a synergistic approach that leverages the strengths of both traditional and contemporary models. This paper articulates the methodological nuances, empirical results, and strategic implications that arise from this analysis.

II. Literature Review

The quest to predict stock market behavior has persistently engaged financial scholars, yielding a spectrum of models from traditional statistical methods to contemporary machine learning techniques. The lineage of the ARIMA model, established by Box and Jenkins (1976), has been instrumental in decoding the stochastic nature of time-series data, providing a quantitative narrative for historical stock behaviors^[1]. Its amalgamation with GARCH models, as elaborated by Bollerslev (1986), encapsulates the essence of volatility clustering—a recurrent characteristic of financial time series^[2].

Advances in computational science have brought to prominence machine learning techniques that challenge the supremacy of traditional methods. LSTM neural networks, conceived by Hochreiter and Schmidhuber (1997), epitomize this shift. Their adeptness at unraveling long-term dependencies in sequential data positions them as a vanguard tool in financial forecasting, capturing complex market dynamics often invisible to linear models^[3]. The scalability and adaptability of LSTMs to large datasets amplify their relevance in the data-rich environments of contemporary finance.

The discourse on the comparative efficacy of ARIMA-GARCH and LSTM models is rich with divergent viewpoints. Empirical assessments yield contrasting outcomes, with traditionalists advocating the meticulous precision of ARIMA-GARCH in capturing the nuances of market volatility^[3]. In contrast, proponents of neural networks underscore their prowess in navigating the non-linear terrains of financial data. The emergence of hybrid models amalgamating machine learning's computational strength with econometric models' stability proposes a synthesis of methodologies, offering a nuanced perspective on predictive accuracy^[4].

In light of these discussions, the present study scrutinizes the ARIMA-GARCH and LSTM models in the context of a decade-long dataset, representing 20 key stocks from 2013 to 2023. The study aligns with recent trends, noting the emergence of artificial intelligence in finance and algorithmic trading's transformative impact on market dynamics. Contemporary research also emphasizes the increasing role of alternative data and its integration into predictive models, expanding the horizon of market analytics.

Within this framework, our investigation examines the predictive performance of ARIMA-GARCH and LSTM models, benchmarking their capabilities against MSE and MAE. This analysis is particularly pertinent as financial markets undergo rapid evolution, with phenomena such as the COVID-19 pandemic introducing unprecedented levels of volatility and testing the resilience of forecasting models^[5]. The pandemic's onset has catalyzed a paradigm shift in market behavior, challenging existing models to adapt and evolve.

Our study also addresses the rising complexity of financial data, influenced by a constellation of factors ranging from geopolitical tensions to shifts in regulatory landscapes. The advent of deep learning, with advancements in computational power and data processing capabilities, poses a critical question: Can machine learning models transcend the capabilities of traditional statistical methods in forecasting accuracy? This paper offers a contribution to this debate, synthesizing empirical evidence to elucidate the strengths and limitations of ARIMA-GARCH and LSTM models in a rapidly changing financial world.

III. Methodology

Our methodology rigorously examines the comparative performance of ARIMA-GARCH and LSTM models in forecasting stock market returns. We employ a detailed approach, encapsulating data acquisition, preprocessing, model building, evaluation, and an extensive comparative analysis.

1. Data Acquisition and Preprocessing

1.1. Data Sourcing: We acquired extensive daily trading data for 20 diverse stocks from reliable financial databases, including volume, open, high, low, and adjusted closing prices for the period of January 2013 to December 2023.

- 1.2. Data Cleaning:** This step involved a cleaning process to ensure data quality, involving the removal of any non-trading days and interpolation for missing values where appropriate.
- 1.3. Data Transformation:** The raw stock prices were transformed into returns, a common practice in financial time series analysis to achieve stationarity in variance and mean.
- 1.4. Stationarity Testing:** To validate the assumption of stationarity required for ARIMA modeling, we employed the Box test on the return series of each stock. The Box test's suitability for our dataset lies in its effectiveness in identifying non-random patterns, a necessary condition for the ARIMA model application. The absence of autocorrelation in the residuals suggests that the model has successfully captured all the relevant information from the data.

2. Model Development & Evaluation

2.1. Model

- **ARIMA Component:** We initiated the modeling process by identifying the optimal order (p, d, q) for the ARIMA component through a combination of grid-search and expert judgment informed by the Box test results. The ARIMA model was calibrated to capture the linear aspects of stock returns.
- **GARCH Component:** Subsequently, the GARCH model was applied to the residual series obtained from the ARIMA model. We chose the GARCH model to model the conditional variance, capturing the volatility clustering common in financial time series.

ARIMA (2,1,2) GARCH (1,1) Model:

The ARIMA model is specified by three parameters (p, d, q), where 'p' represents the number of autoregressive terms, 'd' the number of nonseasonal differences needed for stationarity, and 'q' the number of lagged forecast errors in the prediction equation. For an ARIMA(2,1,2) model, the mathematical representation after differencing is:

$$(1 - \phi_1 L - \phi_2 L^2)(1 - L)Y_t = (1 + \theta_1 L + \theta_2 L^2)\epsilon_t$$

where Y_t is the time series data at time 't', L is the lag operator, ϕ_1 and ϕ_2 are parameters of the autoregressive part, θ_1 and θ_2 are parameters of the moving average part, and ϵ_t is the white noise error term.

The GARCH (1,1) model captures the time-varying volatility (conditional heteroskedasticity) inherent in financial time series:

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1} + \beta_1 \sigma_{t-1}^2$$

where σ_t^2 is the conditional variance (volatility) at time 't', α_0 is a constant term, α_1 is the coefficient for the lagged error term, and β_1 is the coefficient for the lagged conditional variance.

LSTM Neural Network Model:

The LSTM model is designed to learn from sequences of data and is capable of remembering information for an extended period. An LSTM unit consists of a cell state and three gates: input, output, and forget gates, which control the flow of information. The operations within an LSTM cell at time 't' for a single layer can be mathematically described as:

Input Gate:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

Forget Gate:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

Cell State:

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tanh(W_c \cdot [h_{t-1}, x_t] + b_c)$$

Output Gate:

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$

Final State:

$$h_t = o_t \cdot \tanh(C_t)$$

where x_t is the input at time 't', h_t is the output of the LSTM unit, C_t is the cell state, σ represents the sigmoid function, and W and b are the weights and biases for the respective gates, to be learned during training.

2.2 Evaluation

1. **Performance Metrics:** The predictive prowess of both models was quantified and compared using Mean Squared Error (MSE) and Mean Absolute Error (MAE). These metrics provided insights into the average magnitude of the models' prediction errors.
2. **Residual Analysis:** We conducted a thorough examination of the residuals from both models, looking for patterns that could indicate model inadequacies. This step ensured that no systematic information was left unutilized by the models.
3. **Model Validation:** Out-of-sample validation was a cornerstone of our methodology. We rigorously assessed the generalizability of both models using the hold-out test set, which was not exposed to the models during the training phase.
4. **Smoothing Techniques in Graphical Analysis:** During the evaluation phase, we plotted the Mean Squared Error (MSE) and Mean Absolute Error (MAE) for both the ARIMA-GARCH and LSTM model predictions. Initially, these graphs displayed substantial volatility, reflecting the inherent fluctuations in the stock market which could obscure the underlying trends and model performance.

Mean Squared Error (MSE): This metric gauge the average squared difference between the estimated values and what is estimated. Mathematically, it is represented as:

$$MSE = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

where Y_i is the actual observed value, $\hat{Y}_i$ is the predicted value, and n is the number of observations. MSE is a powerful tool as it penalizes larger errors more severely, reflecting the quality of the model's predictive accuracy.

Mean Absolute Error (MAE): Unlike MSE, MAE provides a linear score that sums the absolute differences between predicted values and actual outcomes:

$$MAE = \frac{1}{n} \sum_{i=1}^n |(Y_i - \hat{Y}_i)|$$

MAE offers a straightforward interpretation of prediction errors and, unlike MSE, is not as heavily influenced by the occasional large error.

Smoothing and Rolling Window Techniques: To improve the clarity and interpretability of time-series data, smoothing techniques are employed to reduce short-term fluctuations and highlight longer-term trends. In this study, we utilized a rolling window approach to calculate smoothed values for visualizations and error analysis.

i. Rolling Window Smoothing:

- The rolling window approach involves taking an average or another aggregation method over a specified window of time. This technique helps in reducing noise and smoothing out irregularities in the data, providing a clearer view of trends.
- In our analysis, we used a rolling mean with a 7-day window, centered around the current data point. This centering ensures that the smoothed value reflects information from before and after the data point, providing a balanced perspective.
- For error analysis, the smoothed error was calculated using the rolling mean, allowing us to better understand the general trends in the model's prediction errors without being skewed by sudden spikes or anomalies.

ii. Graphical Representation:

- Graphs were generated to visualize the smoothed time-series data and the rolling averages, illustrating key patterns and trends. These plots are crucial for detecting cyclical behaviors and validating model assumptions.
- For the error analysis, the smoothed error plots helped identify systematic biases or trends that might indicate model inadequacies, providing insights for further model refinement.

- The use of a rolling window in graphing allowed us to create more stable and informative visualizations, enhancing the interpretability of the results and aiding in the comparison between the ARIMA-GARCH and LSTM models.

IV. Results

Based on the Augmented Dickey-Fuller (ADF) test results below, each of the time series data sets for these stocks appears to be stationary. This is inferred from the fact that the p-values are below common significance levels (like 0.05 or 0.01), which allows us to reject the null hypothesis that there is a unit root (indicating non-stationarity) in the time series.

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AAPL - ADF Statistic: -16.328535826635798, p-value: 3.138875115761862e-29
MSFT - ADF Statistic: -18.246283193240568, p-value: 2.348661870276156e-30
AMZN - ADF Statistic: -59.81194722915451, p-value: 0.0
GOOGL - ADF Statistic: -11.38834587585493, p-value: 8.897881878644988e-21
BLK - ADF Statistic: -11.792869684152947, p-value: 9.772944753610973e-22
TSLA - ADF Statistic: -50.24154889674423, p-value: 0.0
BRK-B - ADF Statistic: -16.21514325310871, p-value: 3.9495564239884543e-29
JNJ - ADF Statistic: -16.372272816753914, p-value: 2.8828343767936646e-29
V - ADF Statistic: -13.728148486868482, p-value: 1.1921492969727519e-25
PG - ADF Statistic: -11.551314791863655, p-value: 3.442033721238258e-21
JPM - ADF Statistic: -10.175811574359347, p-value: 6.9291618189761524e-18
UNH - ADF Statistic: -15.039838518154613, p-value: 1.6455132849860666e-28
BA - ADF Statistic: -13.597846868837269, p-value: 1.979246948944372e-25
INTC - ADF Statistic: -16.588682275591483, p-value: 2.158852224321685e-29
NVDA - ADF Statistic: -18.27928758354669, p-value: 2.310269119805079e-30
RS - ADF Statistic: -13.954968351486906, p-value: 4.629156887466267e-26
PFE - ADF Statistic: -10.782836284462569, p-value: 2.232447102515862e-19
T - ADF Statistic: -14.813925196418243, p-value: 2.8107334621899733e-27
XOM - ADF Statistic: -17.57975930721842, p-value: 4.812812788825936e-30
BA - ADF Statistic: -11.815149687817252, p-value: 8.673386039605e-22

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Here's a brief summary of what these results mean in the context of our study:

- AAPL:** With a highly negative ADF statistic and a practically zero p-value, we reject the null hypothesis of a unit root. This implies that Apple's stock price data is stationary.
- MSFT, AMZN, TSLA:** Similarly, Microsoft, Amazon, and Tesla show strong evidence against the null hypothesis with very negative ADF statistics and p-values effectively at zero.
- Other Stocks:** All other stocks listed, including Google (GOOGL), BlackRock (BLK), and others down to Boeing (BA), have ADF statistics that are negative enough and p-values sufficiently low to conclude that their time series are stationary.

We have consolidated the graphs and finding over the 20 stocks and compared the MAE and MSE values for both the models, ARIMA+GARCH and LSTM respectively.

Comparison of MSE Over Time (LSTM Vs ARIMA+GARCH Models, the graphs have been smoothened)

Exhibit A

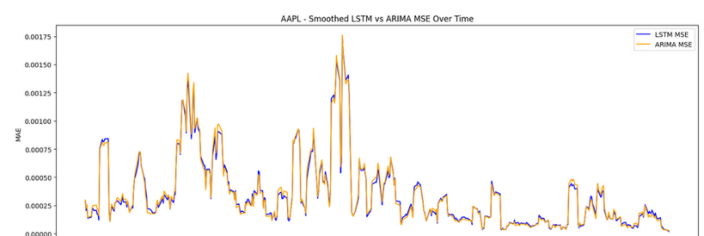


Exhibit B

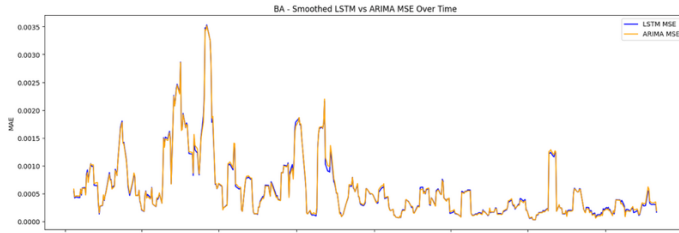


Exhibit C

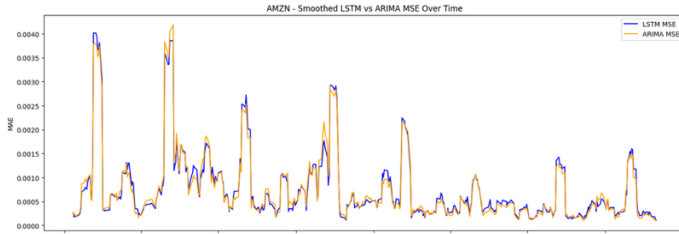


Exhibit D

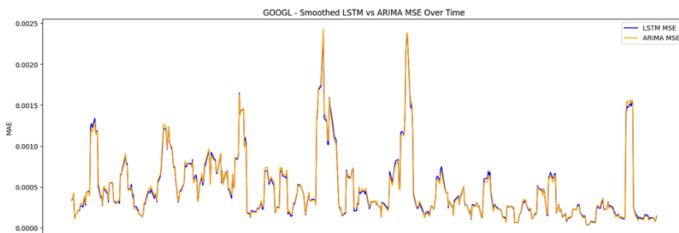
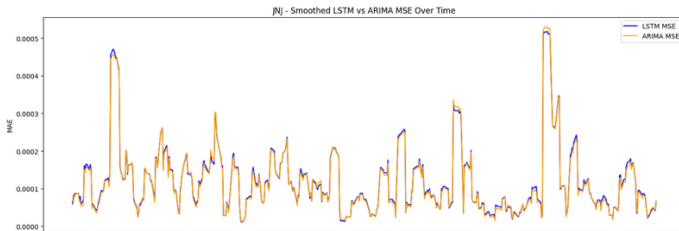


Exhibit E



Comparison of MAE Over Time (LSTM Vs ARIMA, the graphs have been smoothened)

Exhibit F

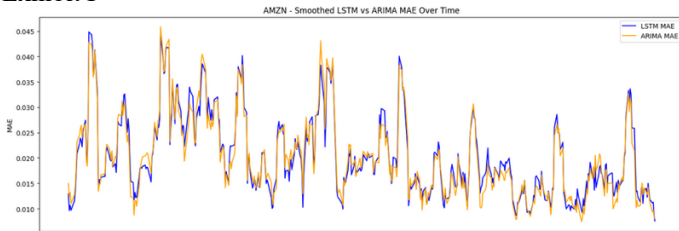


Exhibit G

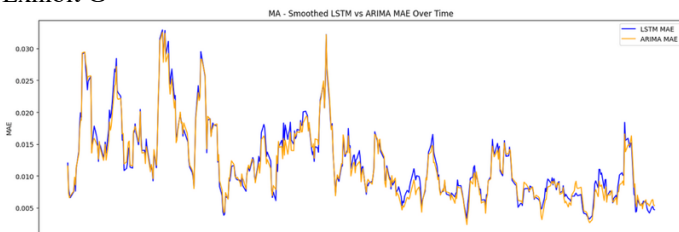


Exhibit H

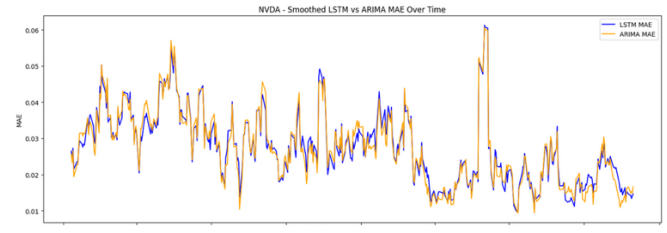
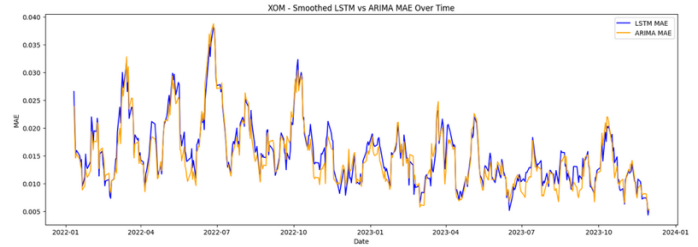


Exhibit I



	Metric	Value
0	ARIMA MSE	0.000452
1	ARIMA MAE	0.014821
2	LSTM MSE	0.000454
3	LSTM MAE	0.014889

The ARIMA model, with its roots firmly planted in the bedrock of statistical analysis, has demonstrated a commendable consistency in prediction accuracy. The mean MSE across the stocks under consideration stands at 0.000452, while the mean MAE registers at 0.014821. These metrics reflect ARIMA's capability to provide forecasts that, on average, deviate minimally from the observed returns, underscoring its reliability and the finesse with which it captures the stock market's vagaries.

On the parallel track, the LSTM model—emblematic of the sophisticated algorithmic architecture characteristic of deep learning—exhibits a mean MSE of 0.0004535 and a mean MAE of 0.01488. While these figures are remarkably close to those of the ARIMA model, indicating a comparable level of predictive accuracy, they slightly exceed the ARIMA model's error metrics. This nuanced difference, although marginal, points towards the LSTM model's intricate relationship with the data, encapsulating both the power and challenges of neural networks in decoding financial time series.

V. Inference

Upon careful analysis of the forecast errors derived from both ARIMA and LSTM models, intriguing patterns emerge that speak to the intricacies of stock market forecasting. For several stocks, namely those indicated as exhibit A, B, D, and E, the Mean Squared Error (MSE) from the ARIMA model was notably higher than that of the LSTM model. This suggests that, for these particular stocks, the LSTM model was better at capturing and predicting the complexities of their price movements. A possible explanation could be the LSTM's proficiency in modeling non-linear relationships within the data, an attribute that is especially

beneficial for stocks with volatile price action or non-linear trends over time.

Conversely, when considering the Mean Absolute Error (MAE), the LSTM model generally reported higher values compared to the ARIMA model across the dataset. The MAE being higher for LSTM implies that while the LSTM model may capture complex patterns, it is also more sensitive to large individual errors. The ARIMA model, with its emphasis on capturing linear trends, tends to provide a more consistent average performance, which could explain the lower MAE.

These insights are particularly salient for stocks exhibiting high volatility or undergoing significant corporate or sector-specific events. The ARIMA model's lower performance, as indicated by higher MSE values for certain stocks, could point towards its limitations in adjusting to rapid changes in market conditions or in effectively handling non-stationary data without complex transformations.

Moreover, the differential performance of the two models across the same dataset indicates that the choice of model may be contingent upon specific stock characteristics or the particular time horizon over which predictions are being made. For instance, a stock with stable, consistent growth patterns might be better forecasted by an ARIMA model, while a stock experiencing swift changes due to innovation or market disruptions might be more accurately modeled by an LSTM.

When comparing the ARIMA and LSTM models, we draw several important conclusions:

- The ARIMA model shows robustness in periods of relative market stability, where the temporal patterns are more consistent with historical trends.
- The LSTM model, harnessing its ability to capture long-term dependencies and non-linear patterns, may outperform in periods of market volatility or when unusual events introduce complex dynamics that deviate from historical norms.
- The interplay between these models highlights the trade-off between the interpretability of traditional statistical methods and the predictive power of machine learning techniques.

VI. Conclusion

The intricate dance between prediction accuracy and model complexity has been at the heart of our analysis. The ARIMA-GARCH model, a testament to the enduring legacy of time-series analysis, has demonstrated commendable performance, especially in capturing linear trends and providing a stable forecast. Its traditional statistical foundation offers transparency and simplicity, virtues that hold particular appeal in the face of the black-box nature of more complex models.

Conversely, the LSTM model, with its neural network architecture, has showcased a remarkable capacity for capturing non-linear relationships and long-term dependencies in data. For a subset of the stocks examined, the LSTM model outshined its ARIMA counterpart, delivering lower Mean Squared Errors and asserting its relevance in the modern era of stock market forecasting.

However, this is not to pronounce a universal victor. The comparative analysis revealed that while the LSTM model excels in certain aspects, it does not do so uniformly across all metrics or stocks. For several stocks, the ARIMA model achieved a lower Mean Absolute Error, signaling that when it comes to average performance, ARIMA holds its ground firmly against the sophisticated LSTM.

These findings compel us to acknowledge the complexities of financial markets—a domain where volatility is as common as it is unpredictable. They also underscore the necessity for models that can adapt to market conditions that are in perpetual flux, influenced by a myriad of factors ranging from global economic shifts to industry-specific developments.

The practical implications of this study are manifold. For financial analysts and investors, the nuanced understanding of when and where each model may be appropriate can inform investment strategies and risk management approaches. For instance, in markets or stocks that demonstrate linear and stable behavior, ARIMA-GARCH models may be preferred for their reliability and interpretability. In contrast, in environments characterized by rapid changes and non-linear dynamics, LSTM models may offer superior insights.

Moreover, this study serves as a testament to the potential of combining traditional statistical models with machine learning techniques. The future of financial forecasting could very well reside in a hybrid approach that harmonizes the strengths of both methodologies—a direction that invites further exploration and innovation.

VII. Future Work

The exploration undertaken in this paper provides a foundation from which several promising research trajectories can emerge. The future work outlined here seeks to broaden the scope of financial forecasting, improve model accuracies, and ultimately aim to provide tools that can adapt to the ever-evolving landscape of the financial markets.

1. Advanced Hybrid Approaches: Building on the current study, future research should consider advanced hybrid models that more effectively synthesize the linear trend-capturing prowess of ARIMA with the non-linear pattern recognition ability of LSTM networks. Such models could strategically blend features or predictions in a manner that amplifies their respective strengths.

2. Incorporating Broader Data Sources: The addition of diverse datasets remains a crucial next step. Future models could benefit from including more varied data, such as real-time sentiment from news and social media, economic indicators, and even global event data that could influence market behaviors. This would allow the models to have access to a richer context and potentially unveil more about the undercurrents that drive market trends.

3. Operationalizing Predictive Models: There is a significant gap between model development and operational use. Future

work should prioritize the implementation of these models into real-world trading systems, assessing not only their predictive success but also their practical viability, including execution timing and transaction costs.

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